

CMP-28 - Self-Model and Constitutional Mirror

Maintain evidence-linked descriptions of HAL identity, capabilities, limitations, governing constraints, and conformance without becoming self-authorizing.

Authority and state ownership

Book I: Decisions 1-7, 24-29, 38-43, 47-51, 56, and 58. Book II: Chapters 30, 33. Authoritative state: Self-Model Records, Constitutional Mirror versions, capability and limitation projections, conformance summaries. Explicitly non-owned: Constitution, constitutional authority, Capability Contracts, certification decisions, source evidence.

Critical invariant

Self-description is descriptive, evidence-linked, and non-self-authorizing; it cannot create identity, authority, or capability.

Logical interfaces

CMP-28-IF-01 / Command / Refresh Self Model

Command `Refresh Self Model` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-28-IF-02 / Command / Publish Constitutional Mirror

Command `Publish Constitutional Mirror` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-28-IF-03 / Command / Invalidate Projection

Command `Invalidate Projection` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-28-IF-04 / Query / Get Self Description

Query `Get Self Description` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-28-IF-05 / Query / Trace Mirror Claim

Query `Trace Mirror Claim` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-28-IF-06 / Query / Get Declared Limitation

Query `Get Declared Limitation` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-28-IF-07 / Event / Self Model Refreshed

Event `Self Model Refreshed` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-28-IF-08 / Event / Mirror Published

Event `Mirror Published` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-28-IF-09 / Event / Projection Invalidated

Event `Projection Invalidated` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Lifecycle, controls, and recovery

Lifecycle: Draft > Verified > Published > Superseded or Withdrawn. All transitions require version, guard, identity, authority where applicable, idempotency, time context, and evidence. Contain: self-issued Authority or Permission, stale capability claim, missing limitation, evidence disconnect, identity fragmentation. Recovery proves identity, state, authority, provenance, dependencies, and invariants before mutation resumes.

Conformance tests

CMP-28-TST-001 - Sole-owner mutation. Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-28-TST-002 - Authority denial. Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-28-TST-003 - Valid lifecycle. Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-28-TST-004 - Invalid lifecycle. Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-28-TST-005 - Critical invariant. Self-description is descriptive, evidence-linked, and non-self-authorizing; it cannot create identity, authority, or capability. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-28-TST-006 - Failure containment. Inject self-issued Authority or Permission, stale capability claim, missing limitation, evidence disconnect, identity fragmentation. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-28-TST-007 - Recovery. Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-28-TST-008 - Contract compatibility. Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-28-TST-009 - Privacy and security. Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-28-TST-010 - Topology independence. Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Traceability and certification

Twelve requirements map to Books I-III and Book X. No open finding or Owner Review item. Final for Book IV v1.0.